EXPERIMENT 06

Automating Business Processes Using Deluge Scripting and Workflows, with Application Testing and Debugging

Patient Appointment Booking Form — Workflow Automation Use Case

SRI JANANI B
241401108

1. Aim

To automate a key business process of the Appointment_Booking_Form application — validating every appointment request and sending a personalised confirmation email the moment a patient successfully submits the form — using Zoho Creator form workflows and Deluge scripts, and to test and debug the automation end-to-end with real form submissions.

This removes the need for front-desk staff to manually check and acknowledge every booking: invalid requests are stopped at submission time, and every valid request is confirmed immediately and consistently.

2. Procedure

Step 1: Open the application in Edit mode — From the Zoho Creator home page, the Appointment_Booking_Form application was opened in Edit mode and the **Workflow** tab was selected, with **Form workflows** chosen.

Step 2: Create the confirmation workflow — A new workflow was created on **Appointment_Booking_Form** with the Record Event set to **Created or Edited** and the Form Event set to **Successful form submission**, so it runs only after a valid record is saved. It was named **Send appointment confirmation mail**.

Step 3: Add a Deluge script action — In **Add New Action**, **Deluge script** was chosen as the action type (Execute without condition), because the message needs dynamic, field-driven content.

Step 4: Write the confirmation script — The script builds the patient name from the Name field, converts the date, time and age to text, joins the selected Existing Conditions, and uses **sendmail** to send an HTML email to **input.Email_Address1** with the subject "Appointment Booking Confirmation – <Department>". **info** statements were added to log each run for debugging.

Step 5: Create the validation workflow — A second workflow, **Validate Appointment Details**, was created on the same form with Record Event **Created or Edited** and Form Event **Validations on form submission**. Its Deluge script uses **alert** and **cancel submit** to block an age outside 1–120 and an appointment date in the past.

Step 6: Review the workflows — Both workflows were saved, and the Form workflows list confirmed they were attached to Appointment_Booking_Form and **Enabled**.

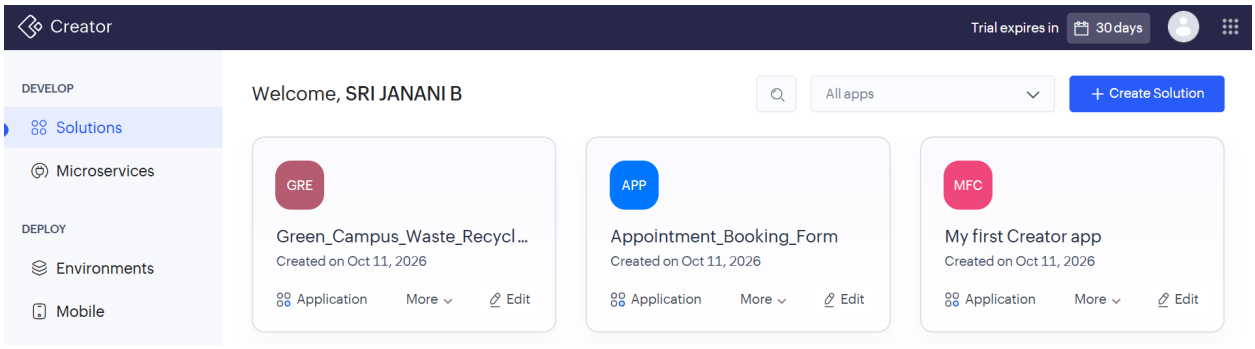
Step 7: Test with a live submission — The application was opened with **Access this application** and a record was submitted: Name SRIJANANI B, Age 20, Gender Female, Department Dermatology, Existing Conditions Hypertension, Preferred Date 31-Oct-2026, Preferred Time 17:00, consent ticked.

Step 8: Debug and verify the output — Errors raised during testing were read from the error dialogs, fixed in the script or account settings, and the form was resubmitted. The patient's inbox was then checked to confirm the email arrived with the correct dynamic values.

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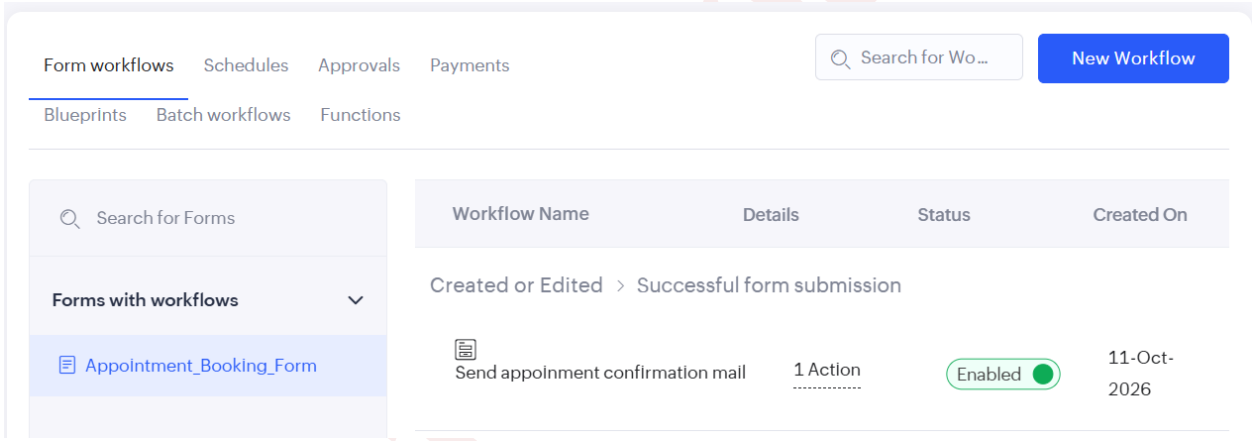
3. Output

3.1 Application in Zoho Creator



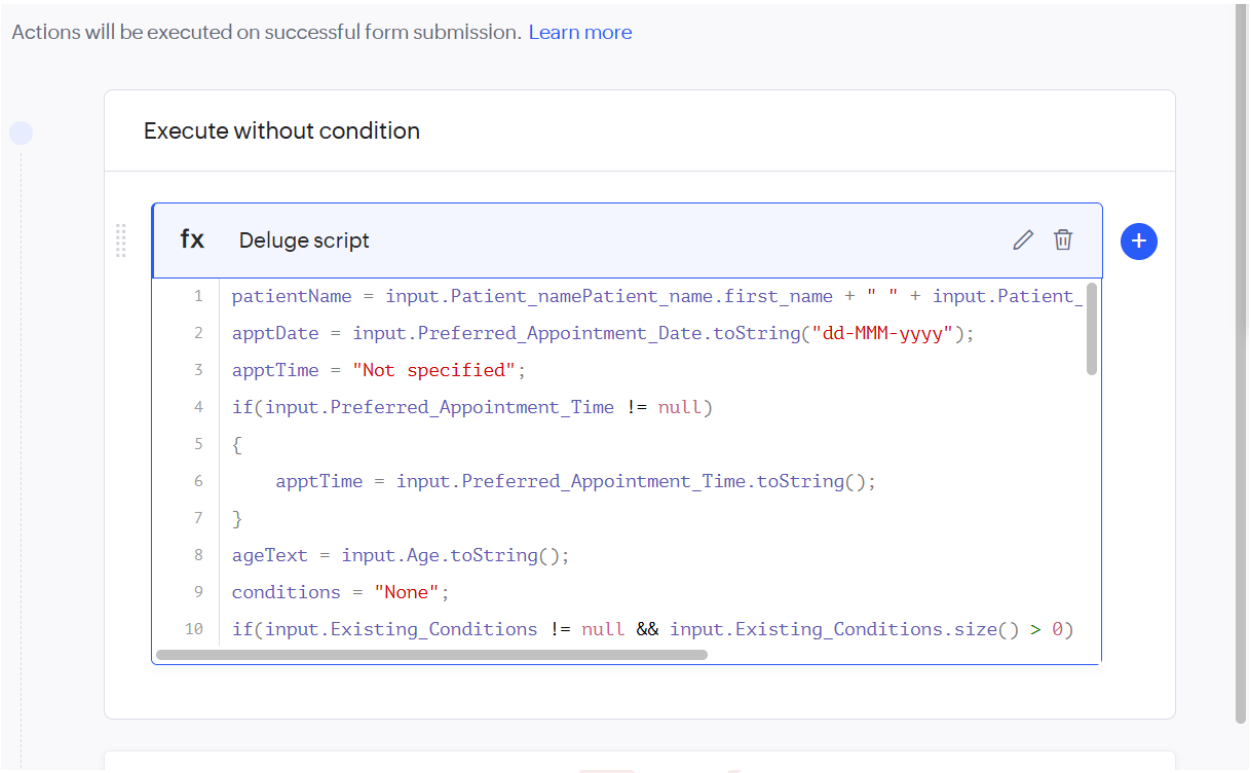
The Appointment_Booking_Form application listed on the Zoho Creator home page, opened in Edit mode to build the workflows.

3.2 First workflow created — trigger configuration



The Form workflows tab shows **Send appointment confirmation mail** attached to Appointment_Booking_Form under **Created or Edited › Successful form submission**, with 1 action and status Enabled.

3.3 Deluge script — confirmation email



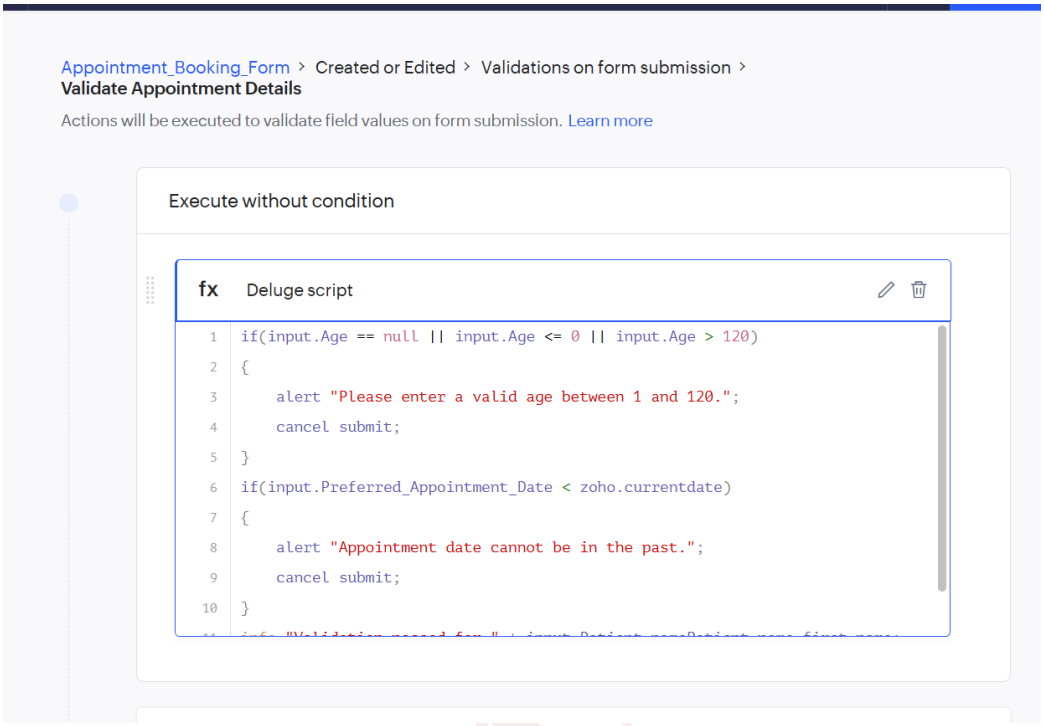
The Deluge script action (Execute without condition). The script converts the date, time and age to text, builds the conditions list and sends the confirmation with **sendmail**.

Full script used:

```
patientName = input.Patient_namePatient_name.first_name + " " + input.Patient_namePatient_name.last_name;
apptDate = input.Preferred_Appointment_Date.toString("dd-MMM-yyyy");
apptTime = "Not specified";
if(input.Preferred_Appointment_Time != null)
{
    apptTime = input.Preferred_Appointment_Time.toString();
}
ageText = input.Age.toString();
conditions = "None";
if(input.Existing_Conditions != null && input.Existing_Conditions.size() > 0)
{
    conditions = input.Existing_Conditions.toString(", ");
}
symptoms = ifnull(input.Symptoms_Description,"-");
info "Workflow started for: " + patientName + " | Email: " + input.Email_Address1;
if(input.Email_Address1 != null && input.Email_Address1 != "")
{
    msg = "<p>Dear " + patientName + ",</p><p>Your appointment request has been received.</p>";
    msg = msg + "<ul><li>Age: " + ageText + "</li><li>Gender: " + input.Gender
              + "</li><li>Department: " + input.Department + "</li>";
    msg = msg + "<li>Preferred Date: " + apptDate + "</li><li>Preferred Time: " + apptTime + "</li>";
    msg = msg + "<li>Existing Conditions: " + conditions + "</li><li>Symptoms: " + symptoms + "</li></ul>";
    msg = msg + "<p>Thank you,<br>Hospital Front Desk</p>";
    sendmail
    [
        from :zoho.loginuserid
        to :input.Email_Address1
        subject :"Appointment Booking Confirmation - " + input.Department
        message :msg
    ]
    info "Confirmation email sent to " + input.Email_Address1;
}
else
{
    info "No email address given, so no email was sent";
```

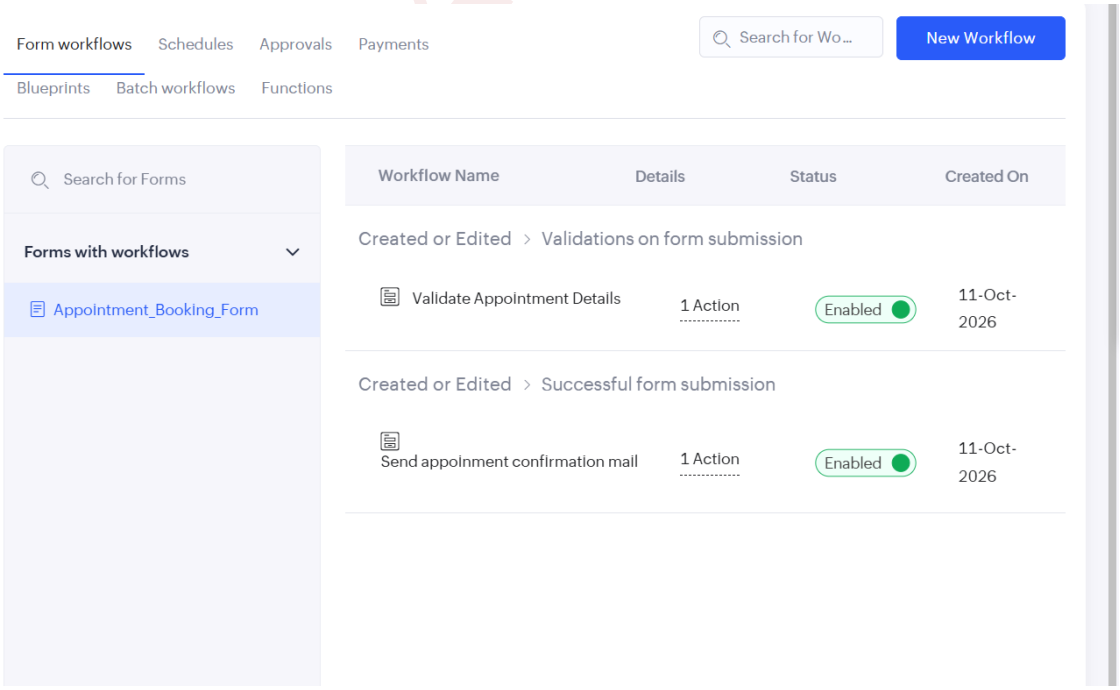
```
}
```

3.4 Deluge script — validation on form submission



The breadcrumb **Appointment_Booking_Form › Created or Edited › Validations on form submission › Validate Appointment Details**. The script rejects an invalid age or a past appointment date with an alert and **cancel submit**, and logs a message with **info** when validation passes.

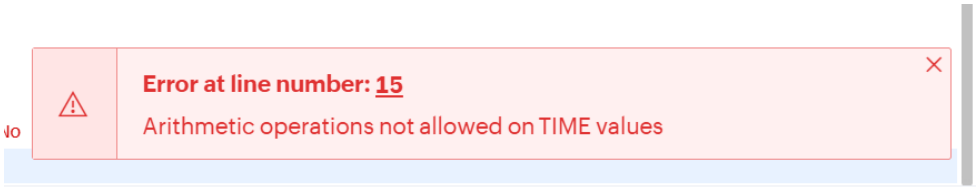
3.5 Saved workflows — summary view



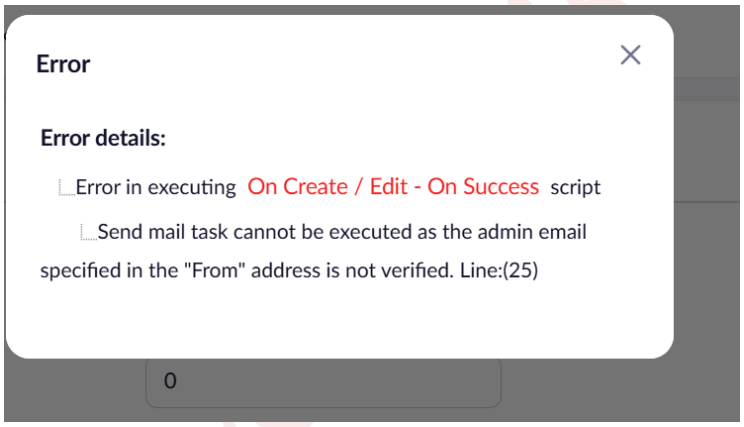
Both workflows on Appointment_Booking_Form are Enabled: **Validate Appointment Details** (Validations on form submission) runs first and can stop the record, and **Send appointment confirmation mail** (Successful form submission) runs only after the record is saved.

3.6 Testing and debugging — errors found and fixed

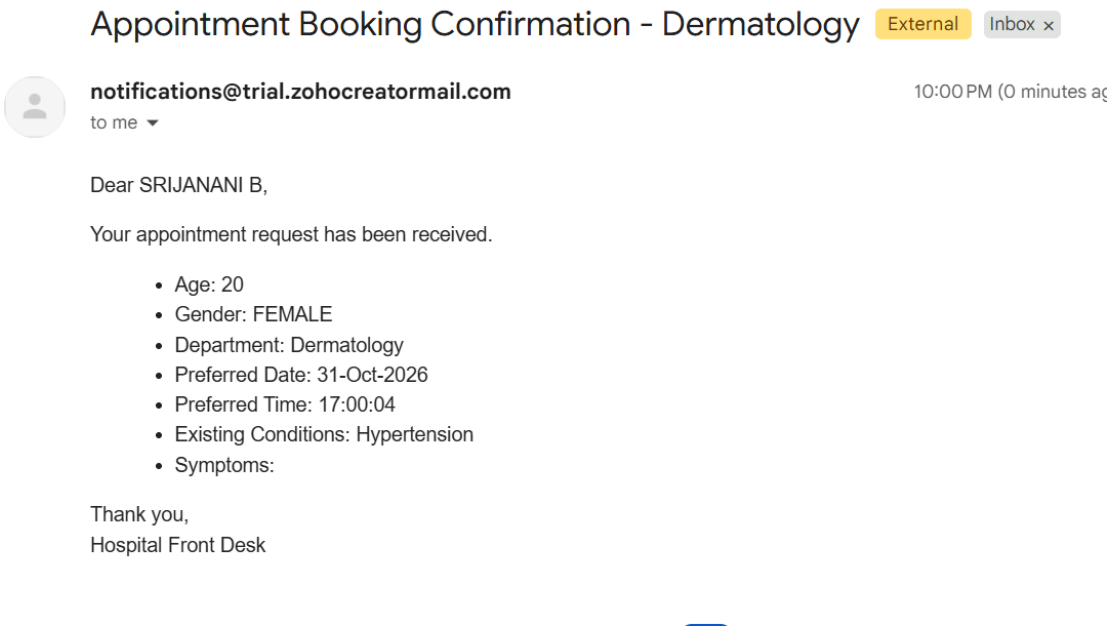
Error 1 — "Arithmetic operations not allowed on TIME values" (line 15). The first version of the script joined the Time field directly into the message string. **Fix:** the time, date and age were converted with **toString()** into text variables before being added to the message.



Error 2 — "Send mail task cannot be executed as the admin email specified in the From address is not verified" (line 25). The record was saved, but the On Success script failed at the sendmail task. **Cause:** the email address on the Zoho account used as the sender was not verified. **Fix:** the From address was set to **zoho.loginuserid** and the account email was verified in Zoho Accounts, after which the workflow ran without errors.



3.7 Debugging output — confirmation email received



The email received after the fixes confirms the automation ran successfully: the subject "Appointment Booking Confirmation – Dermatology" and every dynamic value (Name SRIJANANI B, Age 20, Gender FEMALE, Department Dermatology, Preferred Date 31-Oct-2026, Preferred Time 17:00:04, Existing Conditions Hypertension) were pulled correctly from the submitted record.

4. Final Result

Two form workflows were built on Appointment_Booking_Form using Deluge scripting: **Validate Appointment Details**, which blocks invalid ages and past appointment dates at submission time, and **Send appointment confirmation mail**, which automatically emails a personalised confirmation to the patient after every successful submission.

The automation was tested with live form submissions and debugged using the error messages and **info** logs: a TIME-value concatenation error and an unverified sender-address error were found and fixed, and the final run delivered the confirmation email with all dynamic fields rendered correctly.

The exercise met its objective: notifying patients that their appointment request was received is now fully automated through low-code workflows and Deluge scripting, and testing confirmed the automation behaves reliably end-to-end.

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